

Mathematics for Artificial Intelligence:

A Beginner's Guide

1. LINEAR ALGEBRA



- **Key Concepts:**
- **Vectors** – Lists of numbers representing features (e.g. image pixels, words, inputs)
- **Matrices** – Grids of numbers that store and process large-scale data
- **Matrix Multiplication** – The engine behind neural networks and fast computation
- **Eigenvalues & Eigenvectors** – Detect directions of maximum variance in data

2. CALCULUS



- **Key Concepts:**
- **Derivatives** – Measure how a change in input affects the output
- **Partial Derivatives** – Used when functions have multiple inputs
- **Gradient** – Vector of all partial derivatives (shows direction of change)
- **Gradient Descent** – Optimization method to reduce prediction error
- **Backpropagation** – Core algorithm for training deep learning models using gradients

3. PROBABILITY & STATISTICS



- **Key Concepts:**
- **Probability Distributions** – Models of how likely outcomes are (e.g. normal, binomial)
- **Conditional Probability** – Probability of an event, given another has occurred
- **Bayes' Theorem** – Updating predictions based on new evidence
- **Expected Value** – Average outcome over many trials
- **Confidence Intervals** – Range in which the true value likely falls

4. DISCRETE MATHEMATICS



- **Key Concepts:**
- **Logic & Propositional Statements** – True/false rules for reasoning
- **Boolean Algebra** – Basis for decision-making and binary operations
- **Set Theory** – Grouping, comparing, and filtering data
- **Graphs** – Represent connections (e.g. social networks, maps)
- **Trees** – Hierarchical structures (used in decision trees and parsers)
- **Combinatorics** – Counting and arranging possibilities

INTRODUCTION

Importance of Mathematics in Artificial Intelligence

When people talk about AI, they usually think of computers learning to think, act, and solve problems like humans. But how do computers achieve this? The answer: mathematics!. Mathematics is the language that enables computers to understand patterns, make predictions, and solve complex problems. Without math, AI wouldn't exist. Here's why mathematics is so important for artificial intelligence:

- **Math Helps AI Understand Patterns:** Artificial intelligence is mainly about finding and understanding patterns in data. Math gives AI systems the tools they need to spot these patterns quickly and accurately. For example, when Netflix recommends a show based on your viewing history, it's using math to find patterns in your past choices.
- **Math Helps AI Make Predictions:** Predicting outcomes is another core part of AI. Whether it's weather forecasting, predicting stock market trends, or estimating if a patient might have a disease, math helps AI make accurate predictions based on data.
- **Math Helps AI Improve Continuously:** AI systems are designed to learn from mistakes and get better over time. This learning process relies heavily on math. Mathematical models measure errors and guide the AI to adjust its approach, gradually improving its accuracy.

Do You Need to Be a Math Expert?

While a solid foundation in math is beneficial for anyone working in AI, but you don't need to be a math expert to start your journey. Most modern AI tools (like TensorFlow, PyTorch, and Scikit-learn) handle the hard math for you. So you can build models without deep mathematical knowledge. However, having a basic understanding of key mathematical concepts can significantly enhance your ability to:

- **Interpret Model Performance:** Understanding metrics and statistical analysis helps you evaluate and improve your models.
- **Tweak and Optimize Models:** A grasp of optimization techniques and calculus aids in fine-tuning models for better performance.
- **Diagnose Issues:** When models fail or underperform, knowledge of math helps diagnose problems and implement solutions.

How This Module Helps You ?

Artificial intelligence is not magic—it's math at work. Many people think learning AI is difficult because of maths. You don't need to be a math genius to understand artificial intelligence. With clear explanations, real-world examples, and step-by-step learning, anyone can get a strong grasp of the mathematics behind AI. However, this module breaks down the key mathematical concepts into simple, easy-to-understand with practical explanations

You might be at any age, changing careers, and looking at AI with a mix of excitement and fear. Maybe you haven't touched serious math since school, or you remember only basic arithmetic. This module is written exactly for you.

We assume:

- You can add, subtract, multiply, and divide.
- You are willing to read slowly and think deeply.
- You are ready to learn math to become powerful in the AI era, not just follow tools blindly.

How Is This Module Organized?

The book is built around four pillars of math for AI:

1. **Linear Algebra** – how to work with many numbers at once (vectors and matrices).
2. **Calculus** – how things change and how to optimize them.
3. **Statistics & Probability** – how to understand data and uncertainty.
4. **Discrete Mathematics** – logic, sets, counting, and networks.

Each sub-section follows a pattern to make you understand the concepts more clearly in simple and beginner friendly:

- A short real-life story.
- A clear definition in simple language.
- Important formulas.
- *Why this matters for AI.*
- Few examples.

Chapter 1 – Linear Algebra

Linear algebra is like having organized containers for data. It deals with vectors and matrices, mathematical tools for handling multi-dimensional data.

In AI, many tasks involve vectors and matrices. For example, a single image can be a vector of pixel values. Neural networks use matrices for layers of weights that transform input vectors into output vectors. Operations like dot product measure similarity (used in word embeddings), and matrix multiplication combines linear transformations. These ideas will help us understand how AI algorithms process information in layers and spaces. Hence understanding linear algebra makes it easier to grasp how AI processes information efficiently.

1.1 Scalar and Vector

Imagine you track how many minutes you study AI each day. On Monday you do 30 minutes, Tuesday 45, Wednesday 0, Thursday 60, Friday 40. A single number can't describe your entire week, but a *list* of numbers can. That list is the beginning of **linear algebra**.

- A **scalar** is just a single number, like 5 or -2.3 .
- A **vector** is an ordered list of numbers, like $\mathbf{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$
- A vector is a list of numbers that represents a point or direction in space. For example, is a 2-dimensional vector. We can think of a vector as an arrow from the origin (0,0) to the point (1,2). Vectors have magnitude (length) and direction.

Symbols and Notation

- Scalars: usually lowercase letters like a, b, c .
- Vectors: bold lowercase like $\mathbf{v}$ or with arrows, $\vec{v}$.
- Dimension of a vector: number of entries (e.g., a 5-dimensional vector has 5 numbers).

Why This Matters for AI:

In AI, data are often represented as vectors. For example, an image might be a vector of pixel intensities, and combining data or features means adding or transforming vectors. Understanding vectors lets you understand how AI models handle multi-dimensional inputs.

Worked Examples

Example 1

You drive distances on three days: 10 km, 15 km, 0 km. Represent this as a vector and state its dimension.

Solution:

$$\mathbf{d} = \begin{bmatrix} 10 \\ 15 \\ 0 \end{bmatrix}$$

Dimension=3.

Example 2

You're tracking your monthly expenses in three categories: **rent, food, transport**. For one month, you have

$$\mathbf{e} = \begin{bmatrix} 800 \\ 250 \\ 100 \end{bmatrix}$$

in dollars.

- Which category corresponds to the first component?
- If next month the vector is $\mathbf{e}' = \begin{bmatrix} 800 \\ 300 \\ 120 \end{bmatrix}$, how did your expenses change qualitatively?

Solution:

- First component: rent.
- From $\mathbf{e}$ to $\mathbf{e}'$: food increased (250 $\rightarrow$ 300), transport increased (100 $\rightarrow$ 120), rent stayed the same (800).

1.2 Vector Addition & Scalar Multiplication

You have a weekly study plan:

$$\mathbf{s} = \begin{bmatrix} 30 \\ 45 \\ 0 \\ 60 \\ 40 \end{bmatrix}$$

Your friend has a different plan:

$$\mathbf{f} = \begin{bmatrix} 20 \\ 30 \\ 20 \\ 40 \\ 50 \end{bmatrix}$$

If you study together, what is your combined plan?

- **Vector addition:** Add component-wise. If

$$\mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ \dots \\ a_n \end{bmatrix}, \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \dots \\ b_n \end{bmatrix},$$

then

$$\mathbf{a} + \mathbf{b} = \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \dots \\ a_n + b_n \end{bmatrix}.$$

- **Scalar multiplication:** Multiply each component by the scalar c :

$$c \cdot \mathbf{a} = \begin{bmatrix} ca_1 \\ ca_2 \\ \dots \\ ca_n \end{bmatrix}.$$

Vector addition = “*combine effects.*”

Scalar multiplication = “*stretch or shrink the whole vector.*”

Why This Matters for AI

In AI, vector addition and scalar multiplication are used to combine features, adjust weights, and update parameters in models. Many complex algorithms are built from these simple ideas repeated many times.

Worked Examples

Example 1

Let

$$\mathbf{a} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 1 \\ 4 \\ 0 \end{bmatrix}.$$

Find $\mathbf{a} + \mathbf{b}$.

Solution:

$$\mathbf{a} + \mathbf{b} = \begin{bmatrix} 2 + 1 \\ -1 + 4 \\ 3 + 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix}.$$

Example 2

Let

$$\mathbf{p} = \begin{bmatrix} 1000 \\ 500 \\ 200 \end{bmatrix}$$

be monthly costs (rent, food, transport). What is $0.5\mathbf{p}$?

Solution:

$$0.5\mathbf{p} = \begin{bmatrix} 500 \\ 250 \\ 100 \end{bmatrix}.$$

This could represent half a month’s cost.

1.3 Dot Product and Length (Norm)

Beyond addition and scalar multiplication, vectors have a dot product. For and, the dot product. In general, add the products of corresponding components.

Imagine, You and your friend both track study hours for 3 days. Your vector:

$$\mathbf{y} = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$$

Friend's vector:

$$\mathbf{z} = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$$

How similar are your patterns? One way to compare is the **dot product**.

- **Dot product** of two vectors:

If

$$\mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ \dots \\ a_n \end{bmatrix}, \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \dots \\ b_n \end{bmatrix},$$

then

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + \dots + a_n b_n.$$

- **Length (norm)** of a vector $\mathbf{a}$:

$$\|\mathbf{a}\| = \sqrt{a_1^2 + a_2^2 + \dots + a_n^2}.$$

Dot product = “how much two vectors point in the same direction.”

Norm = “how long the vector is.”

Why This Matters for AI

Dot products and norms help measure similarity and distance between data points. Many AI models (like recommendation systems and word embeddings) rely on dot products to compare users, items, or words in a mathematical space.

Worked Example

Example 1

Let

$$\mathbf{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

Compute:

1. $\mathbf{a} \cdot \mathbf{b}$
2. $\|\mathbf{a}\|$

Solution:

- 1.

$$\mathbf{a} \cdot \mathbf{b} = 1 \cdot 3 + 2 \cdot 4 = 3 + 8 = 11.$$

2.

$$\|\mathbf{a}\| = \sqrt{1^2 + 2^2} = \sqrt{1 + 4} = \sqrt{5}.$$

Example 2

$$\mathbf{y} = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}, \mathbf{z} = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$$

Compute the dot product and interpret.

Solution:

$$\mathbf{y} \cdot \mathbf{z} = 2 \cdot 4 + 3 \cdot 1 + 1 \cdot 2 = 8 + 3 + 2 = 13.$$

1.4 Matrices as Tables / Transformations

Imagine, Your gradebook for 2 students and 3 tests might look like:

	Test 1	Test 2	Test 3
S1	80	90	85
S2	70	75	80

This is a **matrix**. Yes.

- A **matrix** is a rectangular array of numbers.

Example:

$$A = \begin{bmatrix} 80 & 90 & 85 \\ 70 & 75 & 80 \end{bmatrix}$$

- **Rows:** horizontal lines; **columns:** vertical lines.
- Size: 2×3 (2 rows, 3 columns).

Matrices as Transformations

We can think of a matrix as a **machine** that takes a vector in and outputs a new vector.

If A is a suitable matrix and $\mathbf{x}$ is a vector, then

$$A\mathbf{x}$$

represents applying the transformation defined by A to $\mathbf{x}$.

Why This Matters for AI

In AI, data sets are usually stored as matrices: each row can be a data point, each column a feature. Matrices also represent operations like rotations, scaling, and linear layers in neural networks.

Worked Examples

Example 1

Write a 3×2 matrix representing the distances (in km) traveled by 3 days and 2 people:

- Day 1: Person A 5 km, Person B 10 km
- Day 2: Person A 0 km, Person B 3 km
- Day 3: Person A 2 km, Person B 4 km

Solution:

$$D = \begin{bmatrix} 5 & 10 \\ 0 & 3 \\ 2 & 4 \end{bmatrix}$$

Example 2

For

$$A = \begin{bmatrix} 80 & 90 & 85 \\ 70 & 75 & 80 \end{bmatrix},$$

- What is the entry in row 2, column 3?
- How many rows and columns does A have?

Solution:

- Row 2, column 3 entry is 80.
- A has 2 rows, 3 columns $\rightarrow$ size 2×3 .

1.5 Matrix Operations

Matrix addition and scalar multiplication are done element-wise, just like vectors. Example:
Add two 2×2 matrices by adding each corresponding entry.

Matrix multiplication is more complex: If A is $m \times n$ and B is $n \times p$, then AB is an $m \times p$ matrix. The rule: multiply each row of A by each column of B and sum.

Imagine you have a recipe vector (amount of ingredients) and a *cost per unit* vector. The total cost can be written as a matrix times a vector: each row of the matrix corresponds to one recipe, each column to an ingredient.

To multiply $A \cdot B$:

- If A is $m \times n$ and B is $n \times p$, then AB exists and is $m \times p$.
- Entry (i, j) of AB = dot product of row i of A with column j of B .

Matrix multiplication = “row of the first” $\cdot$ “column of the second.”

Why This Matters for AI

In many AI models, the main work is large matrix multiplication: feeding data through layers. The rules you just learned are exactly what computers implement — but at huge scale. This operation helps AI systems like neural networks calculate outputs from inputs.

Worked Example

Example 1

Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}.$$

Compute AB .

Solution:

1. Sizes: Both 2×2 . So AB is 2×2 .

2. Entry (1,1):

$$1 \cdot 5 + 2 \cdot 7 = 5 + 14 = 19.$$

3. Entry (1,2):

$$1 \cdot 6 + 2 \cdot 8 = 6 + 16 = 22.$$

4. Entry (2,1):

$$3 \cdot 5 + 4 \cdot 7 = 15 + 28 = 43.$$

5. Entry (2,2):

$$3 \cdot 6 + 4 \cdot 8 = 18 + 32 = 50.$$

So,

$$AB = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}.$$

Example 2

You have test marks and weightings:

Marks matrix (student $\times$ test):

$$M = \begin{bmatrix} 80 & 90 \\ 70 & 60 \end{bmatrix}$$

Weights vector (importance of each test):

$$\mathbf{w} = \begin{bmatrix} 0.4 \\ 0.6 \end{bmatrix}.$$

Compute final scores as a vector $M\mathbf{w}$.

Solution:

For student 1:

$$0.4 \cdot 80 + 0.6 \cdot 90 = 32 + 54 = 86.$$

For student 2:

$$0.4 \cdot 70 + 0.6 \cdot 60 = 28 + 36 = 64.$$

So

$$M\mathbf{w} = \begin{bmatrix} 86 \\ 64 \end{bmatrix}.$$

1.6 Identity, Inverse and Solving Linear Systems

If you add 5 to a number, subtracting 5 brings you back. Similarly, certain matrices “do nothing” (identity), and some matrices have an “undo” operation (inverse).

Identity Matrix

- The **identity matrix** I behaves like the number 1 for multiplication: $I\mathbf{x} = \mathbf{x}$, $IA = A$, $AI = A$.
- For 2D: $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

Inverse Matrix

- A matrix A is **invertible** if there exists a matrix A^{-1} such that $AA^{-1} = A^{-1}A = I$.
- Not every matrix has an inverse.

Solving Linear Systems

Consider the system:

$$\begin{cases} 2x + y = 5 \\ x - y = 1 \end{cases}$$

We can write this as:

$$A\mathbf{x} = \mathbf{b}, A = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}, \mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}.$$

If A^{-1} exists, then $\mathbf{x} = A^{-1}\mathbf{b}$.

We won't compute inverses by hand in detail; we focus on the idea.

Why This Matters for AI

Many AI tasks involve solving systems of linear equations or approximating such solutions. Understanding the idea of $A\mathbf{x} = \mathbf{b}$ and how matrices can be “inverted” helps you see how models adjust parameters to fit data.

Worked Example

Example

You spend money on **food** and **transport**:

Day 1: 2 food + 1 transport = \$15

Day 2: 1 food + 3 transport = \$17

Find the cost of one unit of food (f) and one unit of transport (t).

System:

$$\begin{cases} 2f + t = 15 \\ f + 3t = 17 \end{cases}$$

Solution:

1. From first: $t = 15 - 2f$.
2. Substitute into second: $f + 3(15 - 2f) = 17$.
3. $f + 45 - 6f = 17 \Rightarrow -5f = -28 \Rightarrow f = 28/5 = 5.6$.
4. Then $t = 15 - 2(5.6) = 15 - 11.2 = 3.8$.

So, food costs \$5.60, transport \$3.80 per unit.

1.7 Eigenvalues & Eigenvectors

Imagine stretching a rubber sheet. Most points move in strange directions, but some lines on the sheet just stretch or shrink without changing direction. Vectors along those lines are like eigenvectors.

Eigenvectors = directions that stay pointing the same way under a transformation.

Eigenvalues = how much they stretch or shrink.

Definitions

For a square matrix A :

- An **eigenvector** $\mathbf{v} \neq \mathbf{0}$ satisfies $A\mathbf{v} = \lambda\mathbf{v}$,
- λ is the corresponding **eigenvalue**.

So A stretches (or flips, or shrinks) $\mathbf{v}$ by factor λ , but does not rotate it to a new direction.

Why This Matters for AI

Eigenvalues/eigenvectors help understand how data spreads and how transformations behave. Methods like PCA (Principal Component Analysis) use eigenvectors to find important directions in data, which is key for dimensionality reduction and visualization.

Example

Consider

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}.$$

- Vector $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$:
 $A\mathbf{v}_1 = \begin{bmatrix} 2 \\ 0 \end{bmatrix} = 2\mathbf{v}_1$. So eigenvalue $\lambda_1 = 2$.
- Vector $\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$:
 $A\mathbf{v}_2 = \begin{bmatrix} 0 \\ 3 \end{bmatrix} = 3\mathbf{v}_2$. So eigenvalue $\lambda_2 = 3$.

Summary

- Vectors are ordered lists of numbers; scalars are single numbers.
- Vector addition and scalar multiplication are the basic operations for combining and scaling vectors.
- Dot product measures similarity; norm measures length.
- Matrices can be seen as tables or machines that transform vectors.
- Identity and inverse matrices are about doing and undoing transformations.
- Eigenvalues and eigenvectors capture special directions that stay aligned under transformations.

What Next ?

Linear algebra helped us **organize and transform** many numbers at once. But in AI, we also care about **how things change** — how loss goes down when we adjust weights, how predictions shift when data changes.

That is where **calculus** comes in:

- Vectors say “here we are.”
- Calculus says “this is how fast things are changing, and in which direction.”

Gentle question: If a model’s error is a number, how can we tell in which direction to move parameters to reduce it?

We use ideas from calculus: **derivatives** and **gradients**.

Chapter 2 – Calculus

Calculus is about change. It studies how functions change with respect to inputs. We focus on two main ideas: derivatives (instant rate of change) and integrals (total accumulation). These are key for understanding optimization in AI.

Calculus lets us optimize functions, which is crucial in AI (e.g. minimizing a loss function). The derivative tells us how to adjust inputs to improve an output. Gradient descent (an AI optimization method) uses derivatives (gradients) to find minima.

2.1 Change and Rates

Imagine, you drive from city A to B. The distance you have traveled depends on time. Your **average speed** over an hour is distance divided by time, but your instantaneous speed at an exact moment is more subtle. Calculus captures this idea.

For a function $f(x)$, the **average rate of change** between $x = a$ and $x = b$ is:

$$\text{Average rate} = \frac{f(b) - f(a)}{b - a}.$$

This is like the slope of the line between two points on the graph.

Why This Matters for AI

Average rates of change are the first step toward derivatives. In AI, we need to know how quickly an error changes as we adjust parameters. That's how models learn.

Worked Example

Example 1

Let $f(x) = 2x$. Find the average rate of change between $x = 1$ and $x = 4$.

Solution:

$$f(1) = 2(1) = 2, f(4) = 2(4) = 8.$$

$$\text{Average rate} = \frac{8 - 2}{4 - 1} = \frac{6}{3} = 2.$$

Example 2

You save money according to $S(t) = 100t$ rupees after t months. Find your average saving rate between 2 and 5 months.

Solution:

$$S(2) = 200, S(5) = 500.$$

$$\text{Average rate} = \frac{500 - 200}{5 - 2} = \frac{300}{3} = 100$$

2.2 Functions & Graphs

Think of a vending machine: you insert a coin (input), press a button, and get a drink (output). A **function** is a math version of this input–output machine.

Definition

A **function** f is a rule that assigns to each input x exactly one output $f(x)$.

- Example: $f(x) = x^2$ squares the input.
- Often, we visualize functions by their **graphs**.

A function's graph is the set of points $(x, f(x))$. For, $f(x) = x^2$ the graph is a parabola opening upward.

Why This Matters for AI

In AI, we use functions to map inputs (features) to outputs (predictions). Understanding how to read and interpret functions and graphs is crucial to understanding model behavior.

Worked Example

Example 1

Let $f(x) = x^2 + 1$. Compute $f(0), f(2), f(-1)$.

Solution:

$$f(0) = 0^2 + 1 = 1, f(2) = 4 + 1 = 5, f(-1) = 1 + 1 = 2.$$

Example 2

Your income in dollars per day is modeled as $I(d) = 50d$ for d working days.

- Find $I(5)$.
- Interpret the meaning in words.

Solution:

$$I(5) = 50 \cdot 5 = 250.$$

Meaning: after 5 working days, you've earned \$250.

2.3 Limits & Continuity

Imagine walking toward a door. You get closer and closer but haven't yet touched it. The distance between you and the door "approaches zero." Limits formalize this idea. The **limit** of $f(x)$ as x approaches a is the value that $f(x)$ gets close to, even if $f(a)$ itself is not defined.

We write:

$$\lim_{x \rightarrow a} f(x) = L.$$

A function is continuous at a if $\lim_{x \rightarrow a} f(x) = f(a)$. That means no jumps or holes at a .

Limits are the foundation of derivatives, which measure instantaneous change. AI learning algorithms use derivatives to decide how to adjust models gradually.

Worked Example

Example 1

Consider $f(x) = 2x$. What is $\lim_{x \rightarrow 3} f(x)$?

Solution:

As x gets close to 3, $f(x) = 2x$ gets close to $2 \cdot 3 = 6$. So

$$\lim_{x \rightarrow 3} 2x = 6.$$

Example 2

Let

$$f(x) = \begin{cases} x^2 & x \neq 1 \\ 100 & x = 1 \end{cases}$$

What is $\lim_{x \rightarrow 1} f(x)$?

Solution:

As x approaches 1 (but not equal to 1), x^2 approaches 1. The function value at 1 is 100, but the limit is still 1. So

$$\lim_{x \rightarrow 1} f(x) = 1.$$

2.4 Derivative

A derivative measures how something changes. Imagine driving a car—your speedometer measures your speed (how fast your distance changes). This is exactly what derivatives do in math—they measure the “rate of change.”

For a function $f(x)$, the **derivative at $x = a$** is: $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$, if this limit exists.

Interpretation: slope of the tangent line at $x = a$.

In AI, derivatives help us understand how changing a parameter (like the weight of a neuron in a neural network) affects the output (like predictions). If changing a parameter significantly improves predictions, we continue adjusting it.

Worked Example

Example 1

For $f(x) = x^2$, find $f'(x)$ and then compute $f'(2)$.

Solution:

Derivative rule: $\frac{d}{dx}(x^2) = 2x$.

So $f'(x) = 2x$. Then $f'(2) = 4$.

Example 2

Suppose your distance (in meters) from home is $d(t) = 4t^2$ after t seconds. Find the instantaneous speed $v(t)$ and then $v(3)$.

Solution:

Derivative: $v(t) = d'(t) = 8t$.

At $t = 3$: $v(3) = 8 \cdot 3 = 24$ m/s.

2.5 Integrals

Let water flows into a tank at a rate that changes over time. To find total water added over a period, you **accumulate** all the small amounts over time — this is integration. Integrals are used to compute expected values, total error, and probabilities. Understanding accumulation helps interpret many AI-related formulas that sum or integrate over data.

Area Under the Curve

The **definite integral** of $f(x)$ from a to b :

$$\int_a^b f(x) dx$$

is interpreted as the area under the curve $y = f(x)$ from $x = a$ to $x = b$ (if $f(x) \geq 0$).

Worked Example

Example 1

Compute $\int_0^2 x dx$.

Solution:

Antiderivative of x is $\frac{1}{2}x^2$.

Evaluate from 0 to 2: $\frac{1}{2}x^2 \Big|_0^2 = \frac{1}{2}(4) - 0 = 2$.

Worked Example

Example 2

If speed is $v(t) = 10$ m/s constant, find distance traveled from $t = 0$ to $t = 5$.

Solution:

Distance: $\int_0^5 10 dt = 10t \Big|_0^5 = 50$

2.6 Gradients and Partial Derivatives

Imagine you stand on a hill and want to go downhill as quickly as possible. You can move in many directions. Which direction reduces your height fastest? The **gradient** tells you.

Partial Derivatives:

For a function of two variables $f(x, y)$:

- The **partial derivative** with respect to x , written $\frac{\partial f}{\partial x}$, treats y as constant and differentiates with respect to x .
- Similarly for $\frac{\partial f}{\partial y}$.

Example:

$$f(x, y) = x^2 + y^2 \Rightarrow \frac{\partial f}{\partial x} = 2x, \frac{\partial f}{\partial y} = 2y.$$

Gradient:

The **gradient** of f is the vector of all partial derivatives:

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix} \text{ (for two variables).}$$

Why This Matters for AI

Gradients are the core of many learning algorithms. When training an AI model, gradients indicate how to change each parameter to reduce error — this is the heart of gradient-based optimization.

Example

For $f(x, y) = x^2 + 3y$, find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

Solution:

$$\frac{\partial f}{\partial x} = 2x, \frac{\partial f}{\partial y} = 3.$$

Summary

- Average rate of change leads to the concept of derivatives, which measure instantaneous change.
- Functions map inputs to outputs; graphs help visualize them.
- Limits formalize approaching behavior and lead to derivatives.
- Integrals represent accumulation (areas, totals).
- Gradients and partial derivatives extend derivatives to many variables.

What Next ?

Calculus gave us tools for understanding **change**. Statistics and probability will give us tools for understanding **variation and uncertainty** in data.

- Calculus: precise change of smooth functions.
- Statistics: patterns and spread in noisy real-world data.
- Probability: models of uncertainty and randomness.

In AI, you almost never have perfect, noiseless data. So you must learn to think in terms of distributions, averages, and chances rather than absolute certainty.

Chapter 3 – Statistics & Probability

3.1 Data Types & Summaries

You ask 5 classmates: “How many hours did you study yesterday?” You get: 0, 1, 2, 2, 4. These numbers are **data**. How do we describe them briefly?

- **Mean (average):** $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$.
- **Median:** middle value when data sorted.
- **Mode:** most frequent value.

Data Types

- **Numerical (quantitative):** e.g., height, hours studied.
- **Categorical (qualitative):** e.g., “Beginner”, “Intermediate”, “Advanced”.

Why This Matters for AI

AI models often start with summary statistics to understand overall levels of features (e.g., average number of clicks per day). Understanding data types and summaries is essential before applying more advanced tools.

Worked Example

Example 1

Data: 0, 1, 2, 2, 4.

- Mean: $\bar{x} = \frac{0+1+2+2+4}{5} = \frac{9}{5} = 1.8$.
- Median (sorted: 0,1,2,2,4) → middle value = 2.
- Mode = 2.

Example 2

Scores: 60, 70, 70, 80.

- Mean: $(60 + 70 + 70 + 80)/4 = 280/4 = 70$.
- Median (sorted: 60,70,70,80) → average of middle two: $(70+70)/2=70$.
- Mode = 70.

3.2 Variance and Standard Deviation

Two classes both have an average test score of 70. But in one class, everyone scored around 70; in the other, some got 40 and some 100. Variance and standard deviation measure spread.

For data $x_1, \dots, x_n$ with mean $\bar{x}$:

- **Variance:**

$$s^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2.$$

- **Standard deviation:**

$$s = \sqrt{s^2}.$$

Why This Matters for AI

Standard deviation tells us how variable data is. Many algorithms assume certain spreads or use standard deviation in loss functions and regularization. Understanding spread helps you detect outliers and noisy data.

Worked Example

Example 1

Data: 1, 2, 3.

1. Mean: $\bar{x} = (1 + 2 + 3)/3 = 2$.
2. Deviations: -1, 0, +1.
3. Squares: 1, 0, 1.
4. Variance: $s^2 = (1 + 0 + 1)/3 = 2/3$.
5. Standard deviation: $s = \sqrt{2/3}$.

Example 2

Data: 0, 1, 2, 2, 4 (from earlier, $\bar{x} = 1.8$).

Compute variance and standard deviation.

Solution:

Deviations: $0 - 1.8 = -1.8$, $1 - 1.8 = -0.8$, $2 - 1.8 = 0.2$, $2 - 1.8 = 0.2$, $4 - 1.8 = 2.2$

Squares: $(-1.8)^2 = 3.24$, $(-0.8)^2 = 0.64$, $(0.2)^2 = 0.04$, $(0.2)^2 = 0.04$, $(2.2)^2 = 4.84$

Sum of squares = $3.24 + 0.64 + 0.04 + 0.04 + 4.84 = 8.8$.

Variance: $s^2 = 8.8/5 = 1.76$.

Standard deviation: $s = \sqrt{1.76} \approx 1.327$.

3.3 Visualizing Data: Histogram, Boxplot, Scatter Plot

A long list of numbers is hard to digest. Visual tools show patterns quickly:

- Are scores clustered?
- Are there outliers?
- Is there a relationship between two variables?

Histogram (Conceptual)

- Break the range of values into **bins** (intervals).
- Count how many data points fall into each bin.
- Draw bars for each bin.

Boxplot (Conceptual)

Shows median, quartiles, and potential outliers. Good for comparing distributions.

Scatter Plot

Plot pairs (x_i, y_i) to see relationships between two numerical variables.

Guided Practice (with Descriptions)

Guided Practice 1

You have heights of 20 people. How would a histogram help?

Solution:

It would show which height ranges are common and whether the distribution is skewed.

Guided Practice 2

You have hours studied vs exam score for 30 students. Why use a scatter plot?

Solution:

To see if higher study hours tend to lead to higher exam scores (positive relationship).

Why This Matters for AI

Visualizing data before modeling helps spot trends, patterns, and problems. Many AI practitioners start with histograms and scatter plots to understand features and targets.

3.4 Probability Basics

If you roll a fair six-sided die, what is the chance of getting a 4? Probability describes how likely events are.

Definitions

- **Sample space:** set of all possible outcomes.
- **Event:** a subset of outcomes.
- **Probability of event A :** $P(A) = \frac{\text{number of outcomes in } A}{\text{total outcomes}}$, assuming all outcomes equally likely.

Basic Rules

- $0 \leq P(A) \leq 1$.
- $P(\text{certain event}) = 1$.
- $P(\text{impossible event}) = 0$.
- For events A and B , $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Why This Matters for AI

Probability is at the heart of AI: models make predictions with some uncertainty.

Understanding probability helps you interpret confidence and risk in model outputs.

Worked Example

Example 1

Roll one fair die. Probability of getting an even number?

Solution:

Even outcomes: {2,4,6} → 3 out of 6.

$$P(\text{even}) = 3/6 = 1/2.$$

Example 2

Suppose a small deck with 4 cards: {A,B,C,D}, each equally likely. Probability of drawing A or B?

Solution:

Favorable outcomes: {A,B} → 2 of 4 = 1/2.

3.5 Conditional Probability & Bayes Idea

You look outside: many people carry umbrellas. That increases the chance that it's raining. That's **conditional probability**: probability given some information.

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)},$$

if $P(B) > 0$.

Interpretation: probability of A given that B has occurred.

Bayes' Idea

Bayes' rule (conceptual form):

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}.$$

It lets us **update beliefs** based on new evidence.

Why This Matters for AI

Bayesian thinking allows models to update beliefs as new data arrives. Many AI systems, especially in recommendation and decision-making, use conditional probabilities.

Example

Suppose 1% of people have a certain condition. A test is positive 99% of the time if a person has it, and 5% of the time if they don't.

Answer:-We won't perform full numeric Bayes here, but the idea: even if the test is positive, your chance of actually having the condition may not be very high, because the condition is rare.

3.6 Random Variables & Expectation

You play a game where you roll a die and win that many dollars. On average, how much money do you expect to win per game? This is expectation.

Random Variable

A **random variable** is a variable whose value depends on the outcome of a random process.

Example: X = number rolled on a fair die.

Expectation (Discrete)

For discrete random variable X with possible values x_i and probabilities p_i :

$$E[X] = \sum_i x_i p_i.$$

Why This Matters for AI

Expectations appear in loss functions, risk measures, and evaluation metrics. Many AI formulas are expectations over data or randomness

Worked Example

Example 1

Fair die: values 1–6, each with probability $1/6$.

$$E[X] = (1 + 2 + 3 + 4 + 5 + 6)/6 = 21/6 = 3.5.$$

Example 2

Game: you win \$10 with probability 0.3, and \$0 with probability 0.7.

$$E[\text{winnings}] = 10 \cdot 0.3 + 0 \cdot 0.7 = 3.$$

3.7 Distributions (Discrete vs Continuous)

Heights of people often follow a “bell-shaped” pattern: most around average, few very short or very tall. This is roughly a **normal distribution**.

Discrete vs Continuous

- Discrete distributions: outcomes are separate (e.g., dice).
- Continuous distributions: outcomes form a continuum (e.g., real-valued heights).

Normal Distribution (High Level)

Normal distribution has parameters:

- Mean μ (center).
- Standard deviation σ (spread).

Most data lies around the mean; symmetric shape.

Why This Matters for AI

Many models assume normality or approximate it. Understanding distributions helps in modeling noise, errors, and uncertainties in AI.

Example 1

Explain the difference between discrete and continuous distributions with an example.

Solution: Dice rolls: discrete (only 1,2,3,4,5,6). Heights: continuous (any real value in range).

Example 2

If a distribution has small σ , what does that mean?

Solution: Data is tightly clustered around the mean.

3.8 Sampling and Simple Inference Ideas

You can't ask every person in a city their opinion. Instead, you ask a **sample** and estimate the overall preference. This is inference.

Sampling

- **Sample:** subset of population.
- Good sampling should be representative (not biased).

Simple Inference

- Use sample mean as an estimate of population mean.
- Larger samples usually give better estimates.

Why This Matters for AI

AI systems are trained on data samples. Understanding sampling and inference helps you judge whether a model trained on a certain dataset will generalize well.

Example 1

Why can sampling be dangerous if biased?

Solution: If the sample is not representative (e.g., only one group), the estimate will be misleading.

Example 2

If two samples have the same mean but one has more data points, which estimate is generally more reliable?

Solution: The one with more data points (larger sample).

Summary

- Descriptive statistics summarize central tendency and spread.
- Variance and standard deviation measure how data is distributed around the mean.
- Visualizations like histograms and scatter plots reveal patterns.
- Probability quantifies uncertainty; conditional probability and Bayes support reasoning with new evidence.
- Random variables, expectation, and distributions are core building blocks for modeling data.

What Next ?

We have explored data, uncertainty, and averages. Now we move to **discrete mathematics**, which deals with:

- Logical statements (true/false),
- Sets of objects,
- Counting,
- Networks and graphs,
- Simple algorithms.

In AI, discrete math helps with **decision logic**, **search**, and **network structures**.

Chapter 4 – Discrete Mathematics

4.1 Basic Logic & Truth (and, or, not)

You set a rule: “I will study if it is NOT a holiday AND I feel healthy.” This involves logical operations.

Propositions and Connectives

- A **proposition** is a statement that is either true (T) or false (F).
- **Not** ($\neg$): flips truth.
- **And** ($\wedge$): true if both are true.
- **Or** ($\vee$): true if at least one is true.

Truth Table Example

For propositions P and Q :

P	Q	$P \wedge Q$	$P \vee Q$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	F

Why This Matters for AI

Logic underlies decision rules, rule-based systems, and reasoning engines. Even learning-based AI must often connect to logical conditions in real applications.

Example 1

If P = “It is raining.” and Q = “I have an umbrella.”

Interpret $P \wedge Q$ in words.

Solution: “It is raining **and** I have an umbrella.”

Example 2

What is $\neg P$ if P is “I am online”?

Solution: “I am not online.”

4.2 Sets and Operations

You have a set of favorite fruits and your friend has another set. You might want to know which fruits you both like.

- A **set** is a collection of distinct objects. Example: $A = \{1,2,3\}$.
- **Union** $A \cup B$: elements in A or B (or both).

- **Intersection** $A \cap B$: elements in both A and B .
- **Difference** $A \setminus B$: elements in A but not in B .

Why This Matters for AI

Sets are foundational in describing collections of states, data points, or options. Many algorithms manipulate sets of candidates, features, or labels.

Example

$A = \{\text{apple, banana}\},$

$B = \{\text{banana, mango}\}.$

- $A \cup B = \{\text{apple, banana, mango}\}.$
- $A \cap B = \{\text{banana}\}.$
- $A \setminus B = \{\text{apple}\}.$

4.3 Counting Principles: Permutations & Combinations

Order matters for passwords: “AB” is different from “BA.”

Order doesn’t matter for lottery numbers: {1,2,3} is same as {3,2,1}.

Basic Formulas

- **Permutations (order matters)**: number of ways to arrange k items from n .

$$P(n, k) = \frac{n!}{(n - k)!}.$$

- **Combinations (order doesn’t matter)**:

$$C(n, k) = \frac{n!}{k! (n - k)!}.$$

Why This Matters for AI

Counting helps evaluate probabilities and understand combinatorial explosion — crucial in search algorithms, state spaces, and model complexity.

Example 1

You have 3 letters: A, B, C. How many 2-letter passwords can you make where order matters and no repetition?

$$P(3, 2) = \frac{3!}{(3 - 2)!} = \frac{6}{1} = 6.$$

They are: AB, AC, BA, BC, CA, CB.

Example 2

Choose 2 fruits from {apple, banana, cherry}. Order doesn’t matter.

$$C(3, 2) = \frac{3!}{2! 1!} = 3.$$

Subsets: {apple,banana}, {apple,cherry}, {banana,cherry}.

Example 3

How many 3-person committees can you form from 5 people?

Solution:

$$C(5,3) = \frac{5!}{3!2!} = 10.$$

Example 4

How many ways to arrange 3 books on a shelf?

Solution:

$$P(3,3) = 3! = 6.$$

4.4 Graphs & Networks

People are **nodes**, friendships are **edges** between nodes. This structure is a **graph**.

- A **graph** consists of **vertices (nodes)** and **edges (links)**.
- An edge may be directed (arrow) or undirected (line).
- Degree of a node: number of edges connected to it.

Why This Matters for AI

Graphs represent social networks, recommendation systems, knowledge graphs, and more. Many AI models are built on graph structures.

Example 1

If a node has edges to 3 other nodes, what is its degree (in an undirected graph)?

Solution: Degree = 3.

Example 2

Give a real-life example of a graph.

Solution: Road network: cities as nodes, roads as edges.

4.5 Modular Arithmetic and Simple Number Properties

On a 12-hour clock, $10 + 5$ hours = 3 (not 15). This is like working “modulo 12”.

Mod Operation

For integers a and $n > 0$,

$$a \bmod n$$

is the remainder when a is divided by n .

Examples:

- $14 \bmod 5 = 4$.
- $13 \bmod 12 = 1$.

Why This Matters for AI

Modular arithmetic appears in hashing functions, randomized algorithms, and optimization over discrete spaces.

Example 1

Compute $20 \bmod 7$.

Solution: $20 = 2 \cdot 7 + 6$, so remainder 6. Result: 6.

Example 2

Why might modular arithmetic be useful in computing?

Solution: It helps in cyclic structures (like clocks), hashing, random number generation, cryptography, and indexing.

4.6 Short Intro to Algorithms & Stepwise Reasoning

An algorithm is like a recipe: a step-by-step procedure to achieve a goal.

- Input $\rightarrow$ process $\rightarrow$ output.
- Each step must be clear and finite.
- Algorithm efficiency (how many steps) matters.

Simple Example: Find Maximum of a List

Steps:

1. Start with first element as current max.
2. Compare with next element; if larger, update max.
3. Repeat until end.
4. Output max.

Why This Matters for AI

Every AI model is implemented using algorithms. Understanding stepwise reasoning helps you understand how models are trained and used in real systems.

Example 1

Describe a simple algorithm to add up a list of 5 numbers.

Solution: Initialize $\text{sum}=0$; add each number in turn; output sum.

Example 2

Why do we care about efficiency in AI algorithms?

Solution: AI often works on huge datasets; inefficient algorithms can take too long or use too much memory.

Summary

- Logic formalizes true/false reasoning with connectives like “and”, “or”, “not”.
- Sets and operations capture collections and their relationships.
- Counting principles underlie probability and combinatorics.
- Graphs represent networks and relations.
- Modular arithmetic and algorithms are basic tools for discrete computation.

Quick Reference Cheat-Sheet

Linear Algebra

- Vector: ordered list $\mathbf{x} = (x_1, \dots, x_n)$.
- Vector addition: $\mathbf{a} + \mathbf{b} = (a_1 + b_1, \dots)$.
- Scalar multiplication: $c\mathbf{a} = (ca_1, \dots)$.
- Dot product: $\mathbf{a} \cdot \mathbf{b} = \sum a_i b_i$.
- Norm: $\|\mathbf{a}\| = \sqrt{\sum a_i^2}$.
- Matrix multiplication: row of first $\cdot$ column of second.
- Eigenvalue equation: $A\mathbf{v} = \lambda\mathbf{v}$.

Calculus

- Average rate of change: $\frac{f(b)-f(a)}{b-a}$.
- Derivative: $f'(x)$, e.g., $\frac{d}{dx}(x^n) = nx^{n-1}$.
- Integral: $\int_a^b f(x) dx$ = area under curve.
- Fundamental theorem: $\int_a^b f(x) dx = F(b) - F(a)$ if $F' = f$.
- Gradient: $\nabla f = (\partial f / \partial x, \partial f / \partial y, \dots)$.

Statistics & Probability

- Mean: $\bar{x} = \frac{1}{n} \sum x_i$.
- Variance: $s^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$.
- Standard deviation: $s = \sqrt{s^2}$.
- Probability: $P(A) \in [0,1]$.
- Conditional: $P(A|B) = P(A \cap B) / P(B)$.
- Expectation: $E[X] = \sum x_i p_i$.

Discrete Math

- Set union: $A \cup B$, intersection: $A \cap B$.
- Permutations: $P(n, k) = n! / (n - k)!$.
- Combinations: $C(n, k) = n! / (k! (n - k)!)$.
- Modulo: $a \bmod n$ = remainder of division.
- Graph: nodes + edges.

High-level AI Uses

- Vectors/matrices for data and weights.
- Derivatives/gradients for learning and optimization.
- Statistics/probability for evaluation and uncertainty.
- Discrete math for logical rules, search, and graph-based models.

THE END